



Quadratic Equation : RACE-01

Single Correct Question

1. The roots of the equation $x^2 - 2x - 1 = 0$
 (A) $\sqrt{2} \pm 1$ (B) $1 \pm \sqrt{2}$ (C) 1, 2 (D) -2, 1
MQE226
2. The real roots x_1 and x_2 of the equation $x^2 + px + 12 = 0$ are such that $x_2 - x_1 = 1$ then value of p -
 (A) ± 5 (B) ± 2 (C) ± 7 (D) ± 6
MQE142
3. Quadratic equation whose roots are the numbers $\frac{1}{10 - \sqrt{72}}$ and $\frac{1}{10 + 6\sqrt{2}}$ is
 (A) $2x^2 - 20x + 1 = 0$ (B) $28x^2 - 20x + 2 = 0$
 (C) $28x^2 - 20x + 1 = 0$ (D) $28x^2 - 20x + 5 = 0$
MQE141
4. If α and β are the roots of $x^2 + px + q = 0$, then the equation whose roots are $(\alpha - \beta)^2$ and $(\alpha + \beta)^2$ is-
 (A) $x^2 - (p^2 - 2q)x + p^2(p^2 - 4q) = 0$ (B) $x^2 - 2(p^2 - 2q)x + p^2(p^2 - 4q) = 0$
 (C) $x^2 - 2(p^2 - 2q)x + p^2(p^2 - q) = 0$ (D) $x^2 - 4(p^2 - 2q)x + p^2(p^2 - 2q) = 0$
MQE167
5. If the sum of the squares of roots of equation $x^2 - 4x + p = 0$ is equal to 16 then the value of p is
 (A) 0 (B) 1 (C) 2 (D) 3
MQE143

Multiple Correct Question

6. If the sum of the roots of the equation $x^2 + (2 - a - a^2)x - a^2 = 0$ equal to zero then the values of a is/are:
 (A) -1 (B) -2 (C) 1 (D) 2
MQE144
7. If the ratio of the roots of the equation $ax^2 - (a + 3)x + 3 = 0$ equal to 1.5 then the values of a is/are:
 (A) 2 (B) -2 (C) $-\frac{9}{2}$ (D) $\frac{9}{2}$
MQE145
8. If the roots of $4x^2 + 5k = (5k + 1)x$ differ by unity, then the values of k is/are
 (A) 3 (B) 5 (C) $-\frac{1}{5}$ (D) $-\frac{1}{3}$
MQE146

Quadratic Equation

Integer Type Question

9. Number of values of x satisfying the equation $\frac{6-x}{x^2-4} = 2 + \frac{x}{x+2}$ is MQE147
10. If α & β are the roots of equation $3x^2 - 5x - 2 = 0$ and if the sum of the cubes of its roots is k , then the value of $27k$ is MQE152

Subjective Type Question

11. Find the roots of equation $x^2 - 8x + 16 = 0$ MQE148
12. Find the roots of the equation $a(x^2 + 1) - (a^2 + 1)x = 0$, where $a \neq 0$. MQE150
13. Express $x_1^3 + x_2^3$ in terms of the coefficients of the equation $x^2 + px + q = 0$, where x_1 and x_2 are the roots of the equation. MQE151

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10
Answer	B	C	C	B	A	B,C	A,D	A,C	1	215

11. 4 12. $a, \frac{1}{a}$ 13. $x_1^3 + x_2^3 = 3pq - p^3$



Quadratic Equation : RACE-02

Single Correct Question

1. If the equation $9x^2 - 2x + a = 6 - ax$ possess equal roots then the value of a is-
 (A) $20 \pm 2\sqrt{5}$ (B) $10 \pm 6\sqrt{5}$ (C) $20 \pm 3\sqrt{5}$ (D) $20 \pm 6\sqrt{5}$ MQE153
2. The values of m for which the equation $x^2 - x + m = 0$ possess no real roots is-
 (A) $m \in (1, +\infty)$ (B) $m \in \left(\frac{1}{2}, +\infty\right)$ (C) $m \in \left(\frac{1}{3}, +\infty\right)$ (D) $m \in \left(\frac{1}{4}, +\infty\right)$ MQE154
3. The value of a for which one root of the equation $x^2 + (2a - 1)x + a^2 + 2 = 0$ is twice as large as the other is -
 (A) -4 (B) -2 (C) -3 (D) -1 MQE155

Multiple Correct Question

4. If the equation $(k - 1)x^2 + (k + 4)x + k + 7 = 0$ has equal roots then the value of k is/are:
 (A) $\frac{-22}{3}$ (B) $\frac{22}{3}$ (C) 2 (D) -2 MQE156
5. If one of the roots of the equation $x^2 - \frac{15}{4}x + a = 0$ is the square of the other then the value of a is/are-
 (A) $-\frac{125}{8}$ (B) $\frac{27}{8}$ (C) $-\frac{27}{8}$ (D) $\frac{125}{8}$ MQE157

Integer Type Question

6. The least integral value of k for which the equation $x^2 - 2(k + 2)x + 12 + k^2 = 0$ has two different real roots is MQE158
7. Number of integer values of c for which the equation $(c - 2)x^2 + 2(c - 2)x + 2 = 0$ possess no real roots is MQE159

Subjective Type Question

8. Nature of roots (real & distinct, real & coincident or imaginary) of equation:
 $x^2 - (7 + \sqrt{3})x + 6(1 + \sqrt{3}) = 0$ MQE160
9. If $2 + \sqrt{3}$ is a root of the equation $x^2 + bx + c = 0$, where $b, c \in \mathbb{Q}$, find b, c . MQE161

10. Consider $f(x) = x^2 + bx + c$.
 (a) Find c if $\alpha, \frac{1}{\alpha}$ are roots of $f(x) = 0$.
 (b) Comment on sign of b & c , if $\alpha < 0 < \beta$ & $|\beta| > |\alpha|$, where α, β are roots of $f(x) = 0$. MQE162
11. If α, β are the roots of the equation $ax^2 + bx + c = 0$, find the value of
 (A) $\frac{1}{\alpha^2} + \frac{1}{\beta^2}$ MQE164
 (B) $\alpha^4\beta^7 + \alpha^7\beta^4$ MQE165
 (C) $\left(\frac{\alpha}{\beta} - \frac{\beta}{\alpha}\right)^2$ MQE166

ANSWER KEY

Question	1	2	3	4	5	6	7
Answer	D	D	A	A,C	A,B	3	2

8. real & distinct. 9. $b = -4, c = 1$
 10. (a) $c = 1$; (b) $b \rightarrow$ negative, $c \rightarrow$ negative
 11. (A) $\frac{b^2 - 2ac}{c^2}$ (B) $\frac{bc^4(3ac - b^2)}{a^7}$ (C) $\frac{b^2(b^2 - 4ac)}{a^2c^2}$



Quadratic Equation : RACE-03

Single Correct Question

1. If α, β are roots of $x^2 - px + q = 0$, then the quadratic equation whose roots are $(\alpha^2 - \beta^2)(\alpha^3 - \beta^3)$ and $\alpha^2\beta^3 + \alpha^3\beta^2$ is-

- (A) $x^2 - p(p^4 - 5p^2q + 5q^2)x + p^2q^2(p^2 - 4q)(p^2 - q) = 0$
 (B) $x^2 - p(p^4 - 5p^2q + q^2)x + p^2q^2(p^2 - 4q)(p^2 - q) = 0$
 (C) $x^2 - p(p^4 - 5p^2q + 5q^2)x + p^2q^2(p^2 - 4q)(p^2 - q) = 0$
 (D) $x^2 - p(p^4 - p^2q + 5q^2)x + p^2q^2(p^2 - 4q)(p^2 - q) = 0$

MQE168

2. If the equation $(a + 3)(a - 1)x^2 + (a - 1)(4a + 3)x + (a^2 - 3a + 2) = 0$ has more than two roots then the value of a is-

- (A) 0 (B) 1 (C) 2 (D) 3

MQE169

Integer Type Question

3. Given two quadratic equations $x^2 - x + m = 0$ and $x^2 - x + 3m = 0$, $m \neq 0$. Find the value of m for which one of the roots of the second equation is equal to double the root of the first equation.

MQE170

Subjective Type Question

4. If $a \cdot \frac{(x-b)(x-c)}{(a-b)(a-c)} + b \cdot \frac{(x-c)(x-a)}{(b-c)(b-a)} + c \cdot \frac{(x-a)(x-b)}{(c-a)(c-b)} = x$ then prove that it is an identity.

MQE171

5. Calculate $\frac{1}{x_1^3} + \frac{1}{x_2^3}$, where x_1 and x_2 are roots of the equation $2x^2 - 3ax - 2 = 0$.

MQE172

6. Find the coefficients of the equation $x^2 + px + q = 0$ such that its roots are equal to p and q .

MQE173

7. If α, β are the roots of $ax^2 + bx + c = 0$, then find the equation whose roots are

(a) $\frac{1}{\alpha^2}, \frac{1}{\beta^2}$

MQE174

(b) $\frac{1}{a\alpha + b}, \frac{1}{a\beta + b}$

MQE175

(c) $\alpha + \frac{1}{\beta}, \beta + \frac{1}{\alpha}$

MQE176

Quadratic Equation

8. For what value of 'a', the equation $(a^2 - a - 2)x^2 + (a^2 - 4)x + (a^2 - 3a + 2) = 0$, will have more than two solutions? Does there exist a real value of 'x' for which the above equation will be an identity in 'a'?

MQE227

ANSWER KEY

- | | | |
|---|--|-------|
| 1. C | 2. B | 3. -2 |
| 5. $\frac{1}{x_1^3} + \frac{1}{x_2^3} = -\frac{27a^3 + 36a}{8}$ | 6. $(p_1 = 0, q_1 = 0, p_2 = 1, q_2 = -2)$ | |
| 7. (a) $c^2x^2 + x(2ac - b^2) + a^2 = 0$
(c) $acx^2 + (a + c)bx + (a + c)^2 = 0$ | (b) $acx^2 - bx + 1 = 0;$ | |
| 8. $(a = 2; \text{No real value of } x)$ | | |



Quadratic Equation : RACE-04

Single Correct Question

1. Number of positive integral values of x satisfying the inequality

$$\frac{(x-4)^{2017} \cdot (x+8)^{2016} (x+1)}{x^{2016} (x-2)^3 \cdot (x+3)^5 \cdot (x-6) (x+9)^{2018}} \leq 0 \text{ is}$$

- (A) 0 (B) 1 (C) 2 (D) 3

MQE228

2. The complete solution of $\frac{x^2-1}{x+3} \geq 0$ & $x^2 - 5x + 2 \leq 0$ is:

(A) $x \in \left[\frac{5-\sqrt{17}}{2}, \frac{5+\sqrt{17}}{2} \right]$

(B) $x \in \left[1, \frac{5+\sqrt{17}}{2} \right]$

(C) $x \in [-3, -1]$

(D) $x \in [-3, -1] \cup [1, \infty)$

MQE229

3. The solution of the inequality $2x - 1 \leq x^2 + 3 \leq x - 1$ is

(A) $x \in \mathbb{R}$

(B) $[-2, 2]$

(C) $(-2, 2)$

(D) $x \in \phi$

MQE230

4. The number of prime numbers satisfying the inequality $\frac{x^2-1}{2x+5} < 3$ is

(A) 1

(B) 2

(C) 3

(D) 4

MQE231

5. The number of the integral solutions of $x^2 + 9 < (x+3)^2 < 8x + 25$ is :

(A) 1

(B) 3

(C) 4

(D) 5

MQE232

6. Number of non-negative integral values of x satisfying the inequality $\frac{2}{x^2-x+1} - \frac{1}{x+1} - \frac{2x-1}{x^3+1} \geq 0$ is

(A) 0

(B) 1

(C) 2

(D) 3

MQE233

Integer Type Question

7. Find the number of integers satisfying the following Inequalities

(i) $x^4 - 5x^2 + 4 \leq 0$

MQE234

(ii) $x^4 - 2x^2 - 63 \leq 0$

MQE235

(iii) $x^2 + 6x - 7 \leq 2$

MQE236

(iv) $\frac{14x}{x+1} < \frac{9x-30}{x-4}$

MQE237

(v) $\frac{x^2+2}{x^2-1} < -2$

MQE238

Quadratic Equation

Subjective Type Question

Solve the following questions for $x \in \mathbb{R}$

8. $\frac{x^2 - 5x + 12}{x^2 - 4x + 5} < 3$

MQE180

9. $\frac{x^2 + 6x - 7}{x^2 + 1} \leq 2$

MQE181

10. $\frac{2x}{x^2 - 9} \leq \frac{1}{x + 2}$

MQE182

11. $\frac{x^2 - 5x + 12}{x^2 - 4x + 5} > 3$

MQE183

12. $\frac{x^2 + 4x + 4}{2x^2 - x - 1} > 0$

MQE184

13. Solve for $x \in \mathbb{R}$

(a) $x^2 - 3x + 4 > 0$

MQE177

(b) $\frac{x}{x^2 + 2} < 1$

MQE178

14. Solve for $x \in \mathbb{R}$

$\frac{(2x - 1)(x + 3)(2 - x)(1 - x)^2}{x^4(x + 6)(x - 9)(2x^2 + 4x + 9)} < 0$

MQE179

ANSWER KEY

Question	1	2	3	4	5	6
Answer	D	B	D	D	D	D

7. (i) 4 (ii) 7 (iii) 9 (iv) 2 (v) 0 8. $x \in \left(-\infty, \frac{1}{2}\right) \cup (3, \infty)$

9. $x \in \mathbb{R}$ 10. $x \in (-\infty, -3) \cup (-2, 3)$

11. $\left(\frac{1}{2}, 3\right)$ 12. $(-\infty, -2) \cup \left(-2, -\frac{1}{2}\right) \cup (1, \infty)$ 13. (a) $x \in \mathbb{R}$; (b) $x \in \mathbb{R}$

14. $x \in (-6, -3) \cup \left(\frac{1}{2}, 2\right) - \{1\} \cup (9, \infty)$



Quadratic Equation : RACE-05

Single Correct Question

1. If the inequality $x^2 - (k - 3)x - k + 6 > 0$ is valid for all real x then the values of k is-
 (A) $(3, -5)$ (B) $(-3, 5)$ (C) $(3, 5)$ (D) $(-3, -5)$ MQE194
2. Find the range of $\frac{x^2 + 2x + 1}{x^2 + 2x + 7}$, where x is real
 (A) $[0, 1)$ (B) $[0, 2)$ (C) $[1, 3)$ (D) $[1, 2)$ MQE195
3. If x is real, then the range of $\frac{x^2 - x + 1}{x^2 + x + 1}$ is-
 (A) $[1, 3]$ (B) $[1, 2]$ (C) $\left[\frac{1}{3}, 3\right]$ (D) $\left[\frac{1}{2}, 2\right]$ MQE196
4. If x be real then the range of $\frac{x}{x^2 - 5x + 9}$ is-
 (A) $\left[-\frac{1}{10}, 1\right]$ (B) $[-1, 1]$ (C) $[-1, 3]$ (D) $\left[-\frac{1}{11}, 1\right]$ MQE197

Multiple Correct Question

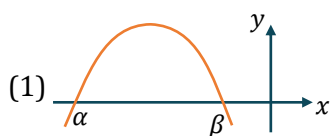
5. If $x^2 + bx + c = 0$ & $2x^2 + 9x + 10 = 0$ have both roots in common then the correct statement is/are-
 (A) $b = \frac{9}{2}$ (B) $c = 3$ (C) $b = \frac{7}{2}$ (D) $c = 5$ MQE198
6. If $x^2 - 7x + 10 = 0$ & $x^2 - 5x + c = 0$ have a common root then the values of c is/are-
 (A) 1 (B) 0 (C) 6 (D) 2 MQE199

Integer Type Question

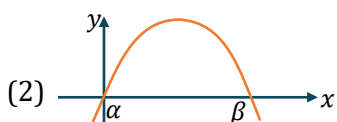
7. The absolute value of a for which the equations $x^2 + ax + 1 = 0$ and $x^2 + x + a = 0$ have exactly one root in common is MQE200
8. Find the sum of the minimum value of $y_1 = x^2 + 2x + 2$ and $y_2 = 4x^2 - 16x + 15$ MQE201
9. For what integral k is the inequality $x^2 - 2(4k - 1)x + 15k^2 - 2k - 7 > 0$ valid for any real x ? MQE202
10. For what least integral k is the expression $(k - 2)x^2 + 8x + k + 4 > 0$ for all values of x ? MQE203
11. If the greatest value of $\frac{x+2}{2x^2+3x+6}$ for real values of x is $\frac{k}{3}$, then the value of k is MQE204

Subjective Type Question

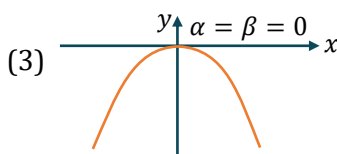
12. For following graphs of $y = ax^2 + bx + c$ with $a, b, c \in \mathbb{R}$, comment on the sign of :
 (i) a (ii) b (iii) c (iv) D (v) $\alpha + \beta$ (vi) $\alpha\beta$



MQE205



MQE206



MQE207

13. Given the roots of equation $ax^2 + bx + c = 0$ are real & distinct, where $a, b, c \in \mathbb{R}^+$, then the vertex of the graph will lie in which quadrant.

MQE208

14. Find the range of 'a' for which :

(a) $ax^2 + 3x + 4 > 0 \quad \forall x \in \mathbb{R}$

MQE209

(b) $ax^2 + 4x - 2 < 0 \quad \forall x \in \mathbb{R}$

MQE210

15. The trinomial $ax^2 + bx + c$ has no real roots, $a + b + c < 0$. Find the sign of the number c .

MQE211

ANSWER KEY

Question	1	2	3	4	5	6	7	8	9	10	11
Answer	B	A	C	D	A,D	B,C	2	0	3	5	1

12. (1) (i) $a < 0$ (ii) $b < 0$ (iii) $c < 0$ (iv) $D > 0$ (v) $\alpha + \beta < 0$ (vi) $\alpha\beta > 0$
 (2) (i) $a < 0$ (ii) $b > 0$ (iii) $c = 0$ (iv) $D > 0$ (v) $\alpha + \beta > 0$ (vi) $\alpha\beta = 0$
 (3) (i) $a < 0$ (ii) $b = 0$ (iii) $c = 0$ (iv) $D = 0$ (v) $\alpha + \beta = 0$ (vi) $\alpha\beta = 0$
13. Third quadrant 14. (a) $a > 9/16$ (b) $a < -2$ 15. $c < 0$



Quadratic Equation : RACE-06

Single Correct Question

1. If α, β are roots of $7x^2 + 9x - 2 = 0$ where $\alpha < \beta$ then select the incorrect statement -
 (A) α and β are greater than -3
 (B) '0' lies in between the roots α and β .
 (C) α and β are less than 1.
 (D) α and β are greater than 1.

MQE212

2. If $a > 1$, roots of the equation $(1 - a)x^2 + 3ax - 1 = 0$ are -
 (A) one positive one negative
 (B) both negative
 (C) both positive
 (D) both non-real

MQE213

3. If the roots of the equation $x^2 - 2ax + a^2 + a - 3 = 0$ are less than 3 then find the values of a is-
 (A) $a < 2$ (B) $a > 2$ (C) $a < 3$ (D) $a > 3$

MQE214

4. If α, β are the roots of $x^2 - 3x + a = 0$, $a \in \mathbb{R}$ and $\alpha < 1 < \beta$, then find the values of a is-
 (A) $a < 2$ (B) $a > 2$ (C) $a < 3$ (D) $a > 3$

MQE215

5. If two roots of $x^3 + 3x^2 - 9x + c = 0$ are equal, then the values of c is-
 (A) $-28, 3$ (B) $-21, 2$ (C) $-27, 5$ (D) $-17, 2$

MQE216

Integer Type Question

6. If α, β are roots of $4x^2 - 16x + \lambda = 0$, $\lambda \in \mathbb{R}$ such that $1 < \alpha < 2$ and $2 < \beta < 3$, then number of integers point in the range of λ .

MQE217

7. If roots of the equation $(2 - x)(x + 1) = a$ are real and positive for $a \in \left(k, \frac{l}{4}\right]$ then the value of $l - k$ is

MQE218

Subjective Type Question

8. If x and y are two real quantities connected by the equation $9x^2 + 2xy + y^2 - 92x - 20y + 244 = 0$, then show that x will lie between 3 and 6. and y between 1 and 10.

MQE219

9. Let α, β be two of the roots of the equation $x^3 - px^2 + qx - r = 0$. If $\alpha + \beta = 0$, then show that $pq = r$

MQE220

10. If α, β, γ be the roots of $ax^3 + bx^2 + cx + d = 0$, then find the value of

(a) $\sum \alpha^2$

MQE221

(b) $\sum \frac{1}{\alpha}$

MQE222

(c) $\sum \alpha^2(\beta + \gamma)$

MQE223

ANSWER KEY

Question	1	2	3	4	5	6	7
Answer	D	C	A	A	C	3	7

10. (a) $\frac{1}{a^2}(b^2 - 2ac)$ (b) $-\frac{c}{d}$ (c) $\frac{1}{a^2}(3ad - bc)$